Building a MOUNTAIN CHAIN

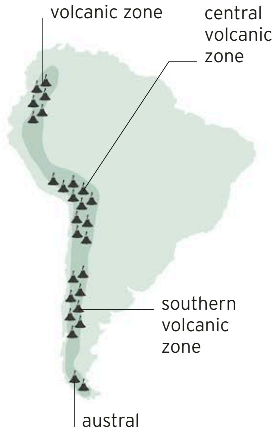
Most mountain chains form where two tectonic plates, at least one of which carries continental crust, move toward each other, or converge, driven by convection in Earth's mantle. If both carry landmasses, these collide and mountains are pushed up; the Himalayas formed in this way. But a mountain chain can also form if there is a landmass on just one of the plates—the formation of the Andes provides a prime example.

The creation of the Andes

The process that built the Andes unfolded over tens of millions of years and began with the development of a new convergent plate boundary to the west of South America. The ocean-dominated plates to the west of the boundary, which include the precursor of today's Nazca and Cocos Plates, as well as the Antarctic Plate, began to subduct, or move under, the South American Plate to their east. This movement squeezed the lithosphere (the crust and top layer of mantle) of the western part of South America, causing major fracturing, called thrust faulting, and uplift of blocks of lithosphere. In parts of the region, a single mountain chain was created; in others, parallel subranges were pushed up in different phases. The subduction process led to the production of magma at depth, which rose up to create volcanism in many parts of the chain.

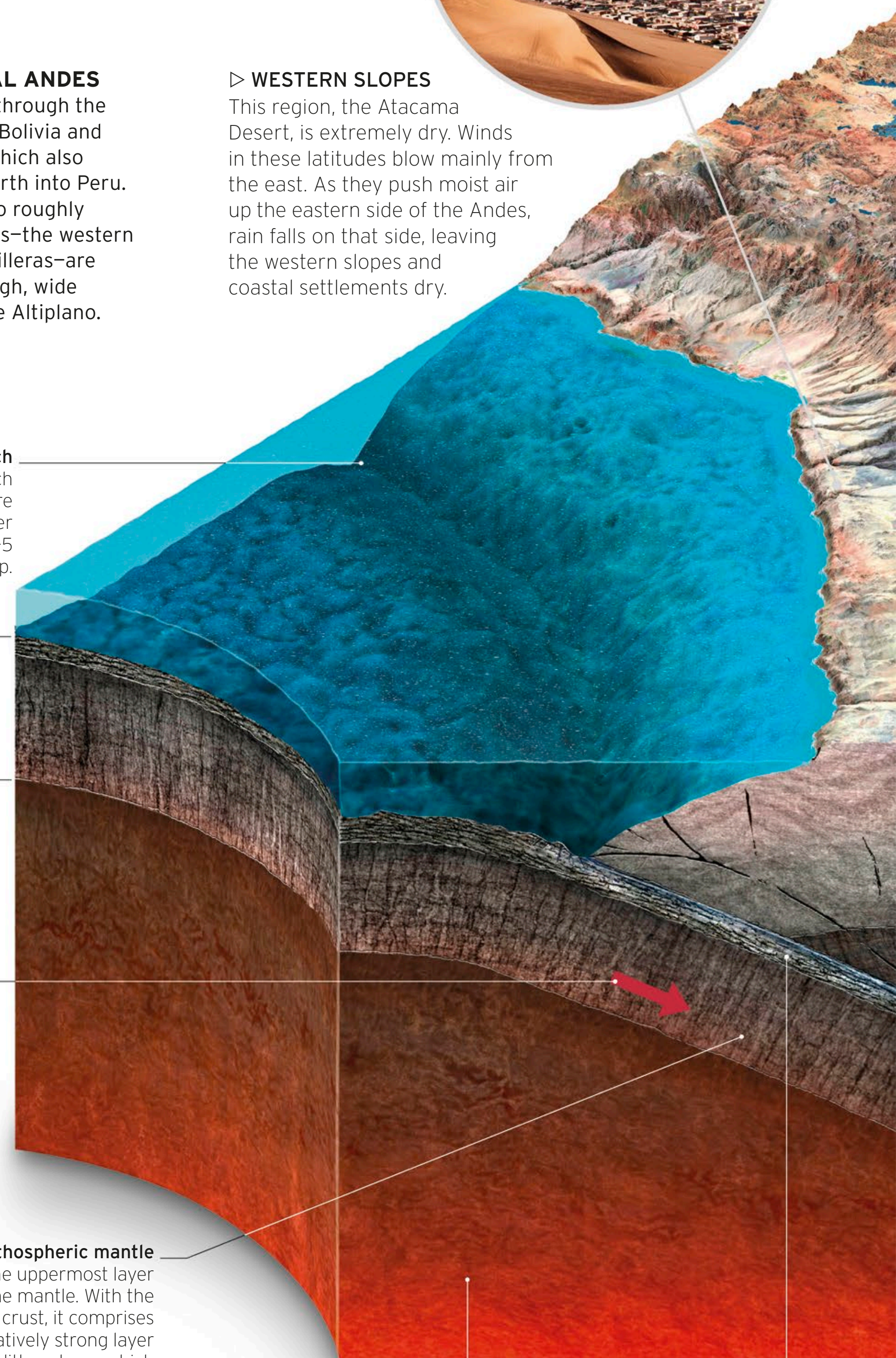
ANDEAN VOLCANIC ZONES

The Andean volcanoes are clustered in four regions—the northern, central, southern, and austral volcanic zones. Between these belts lie large gaps containing no volcanoes. For example, there are around 30 volcanoes in southern Peru, but none in the rest of the Peruvian Andes. The reasons for these gaps are not fully known. One theory is that the tectonic plates that subduct under South America do so at a narrower angle of dip in these "gap" regions.



▶ THE CENTRAL ANDES
This is a section through the central Andes of Bolivia and northern Chile, which also extends in the north into Peru. In this region, two roughly parallel subranges—the western and eastern cordilleras—are separated by a high, wide plateau called the Altiplano.

▷ WESTERN SLOPES
This region, the Atacama Desert, is extremely dry. Winds in these latitudes blow mainly from the east. As they push moist air up the eastern side of the Andes, rain falls on that side, leaving the western slopes and coastal settlements dry.



As the Andes formed, **South America shortened** from west to east